

Evaluation Report & Next-Step Execution Roadmap

Intern Name: Suryansh Raj

Assigned Construct: Working Memory (Gsm)

Proposed Task: Classroom Scenario Recall

Evaluation Status: Proceed with Modifications and Controlled Memory Load

1. Executive Summary

The proposed Classroom Scenario Recall task demonstrates excellent alignment with our vision of building an AI-driven cognitive assessment platform that measures working memory through engaging and ecologically valid experiences. The integration of audio instructions, visual grids, distractors, and delayed recall creates a rich environment for measuring multiple components of Working Memory. The task is adaptive, difficult to fake, and capable of generating valuable behavioral analytics. However, if scenarios become too content-heavy or language-dependent, the task may begin measuring Long-Term Comprehension and Language Ability instead of Working Memory. To align with our startup objectives, the task should maintain minimal language complexity, controlled information load, and emphasize memory updating, attention control, and interference management.

2. Alignment with MentiScope's Objectives

- Alignment with CHC-Gsm Theory: 9.5/10
- Scientific Validity: 9/10
- Innovation: 9/10
- Adaptive Potential: 9.5/10
- AI Readiness: 9.5/10
- MVP Feasibility: 9/10
- Overall Evaluation: 9.2/10 – Proceed with Modifications

3. Scientific Assessment

Primary Working Memory components to measure:

- Attention Control
- Memory Updating
- Sequential Processing
- Cognitive Control
- Interference Management

Potential construct contamination:

- Long-Term Comprehension
- Language Ability
- Prior Knowledge and Classroom Experience
- Reading Comprehension

The task should primarily assess temporary storage and active manipulation of information while minimizing dependence on language comprehension and academic knowledge.

4. Recommended Task Architecture

Module 1 – Auditory Recall

Students hear simple instructions containing 5–7 information chunks and recall key details.

Measures: Auditory Working Memory and attention control.

Module 2 – Visual Grid Memory

Students observe visual grids and recall positions after a delay.

Measures: Visual Working Memory and memory updating.

Module 3 – Distractor Challenge

Introduce secondary tasks during retention intervals.

Measures: Interference susceptibility and cognitive control.

Module 4 – Sequential Recall

Students recall information in correct order and update memory with new instructions.

Measures: Sequential processing and working memory updating.

5. Recommended Scoring Model

- Auditory Memory Score – 25%
- Visual Memory Score – 25%
- Sequential Recall Score – 25%
- Interference Resistance Score – 25%

6. AI Analytics to Capture

- Recall accuracy
- Visual memory performance
- Auditory memory performance
- Serial order accuracy
- Interference susceptibility
- Memory updating efficiency

- Response time
- Error patterns
- Attention stability
- Learning curve

7. MVP Deliverables (Current Phase)

- ✓ Audio instruction generator
- ✓ Visual grid generator
- ✓ Distractor module
- ✓ Item randomization engine
- ✓ Memory score dashboard
- ✓ Event logger
- ✓ Analytics dashboard
- ✓ Documentation
- ✓ Pilot dataset

8. Assessment SDK Compliance Requirements

All outputs must follow the common Assessment SDK specification.

Session Data:

student_id, session_id, construct, task_id, module_version, difficulty_level, start_time, end_time, timestamp.

Event Data:

event_id, student_id, session_id, construct, task_id, item_id, event_type, response, correct, reaction_time_ms, error_type, difficulty_level, timestamp.

Score Output:

raw_score, normalized_score, percentile, sub_scores, confidence_score.

Analytics Output:

behavioral_metrics and recommendations.

9. Next Sprint Execution Plan (2 Weeks)

Sprint Objective:

Refine the task architecture to improve construct purity and develop the foundational MVP

prototype.

Expected Deliverables:

1. Literature review on Working Memory (Gsm) and CHC theory.
2. Construct map defining attention control, memory updating, sequential processing, and cognitive control.
3. Design specifications for all four modules.
4. Audio instruction generator prototype.
5. Visual grid generator prototype.
6. Distractor module specifications.
7. Event logging implementation.
8. Demonstration prototype of one memory assessment module.

10. Definition of Done (DoD)

Scientific:

- ✓ Working Memory components clearly defined.
- ✓ Language complexity controlled.
- ✓ Information load calibrated to 5–7 chunks.

Technical:

- ✓ Audio generator implemented.
- ✓ Visual grid generator functional.
- ✓ Distractor module operational.
- ✓ Event logging and analytics dashboard available.
- ✓ Results exported in SDK format.

Validation:

- ✓ Internal consistency ≥ 0.80 .
- ✓ Pilot study with at least 100 students.
- ✓ Preliminary evidence of working memory measurement.
- ✓ Correlation studies with Attention and Processing Speed.

Guidance to the Intern

Your responsibility is not to build the entire platform at this stage. Your responsibility is to build a scientifically valid micro-assessment engine for Working Memory that follows the common Assessment SDK. Focus on creating adaptive memory tasks, controlled cognitive load, and rich behavioral analytics. This module will later be integrated into the unified cognitive and career assessment platform.